

Git Refresher

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09 October, 2026

Intro

1. What is git?
2. How does git work?
3. Collaboration strategy
4. New SOP

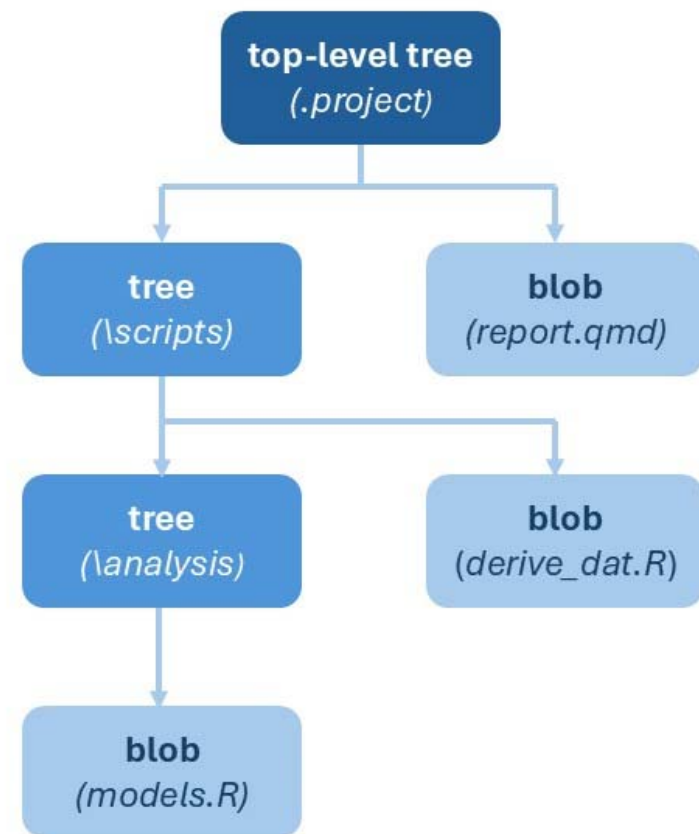
1. What is git?

- Git $\neq$ GitHub
- Git is a version control system (software)
- Github is an online repository hosting platform (server)

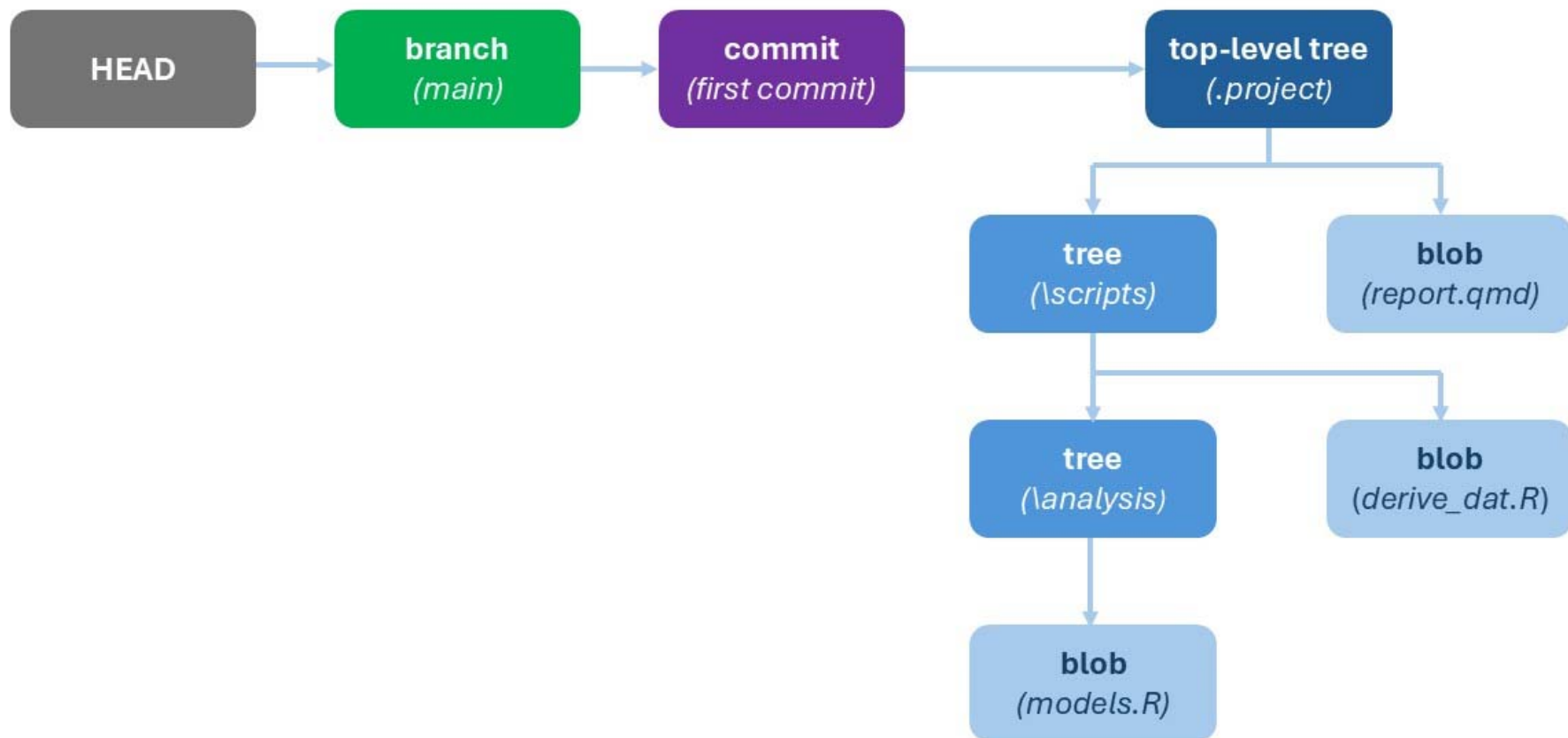
2. How does git work?

2.1 Git data model

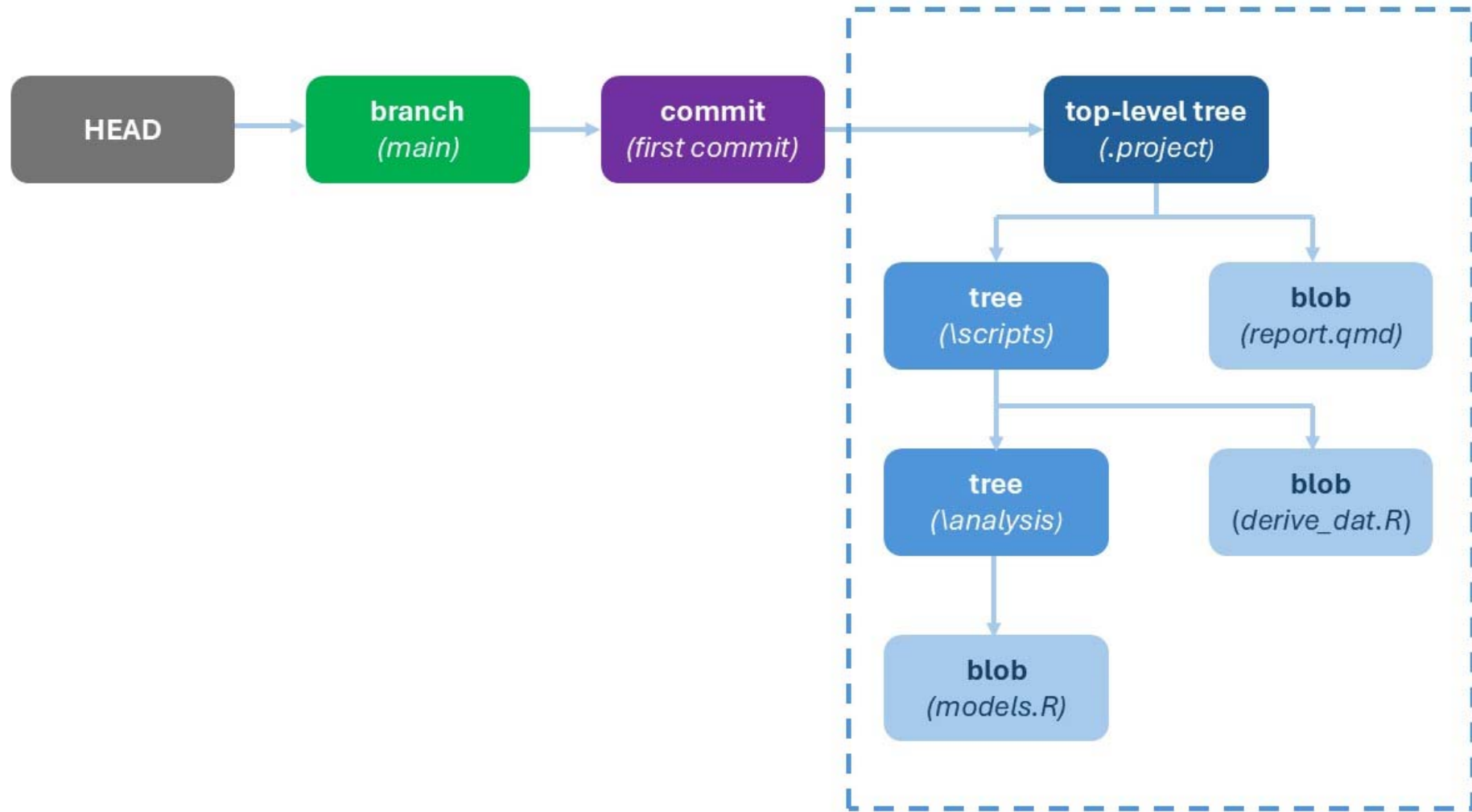
2.1 Git data model



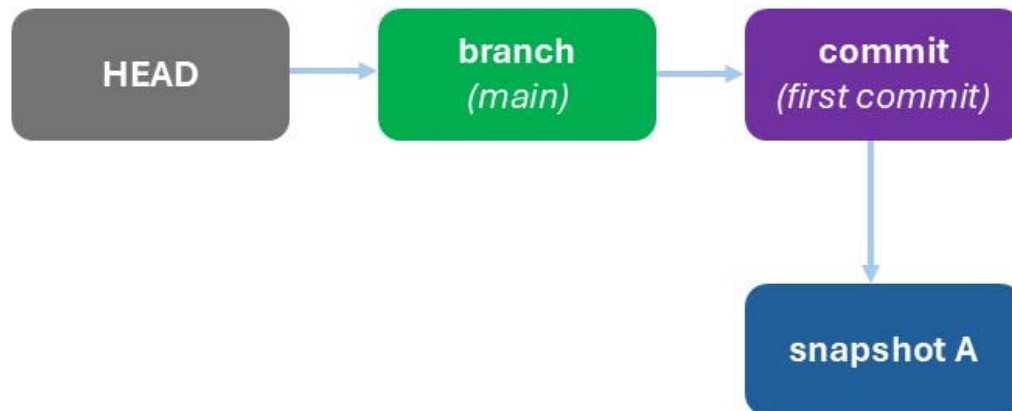
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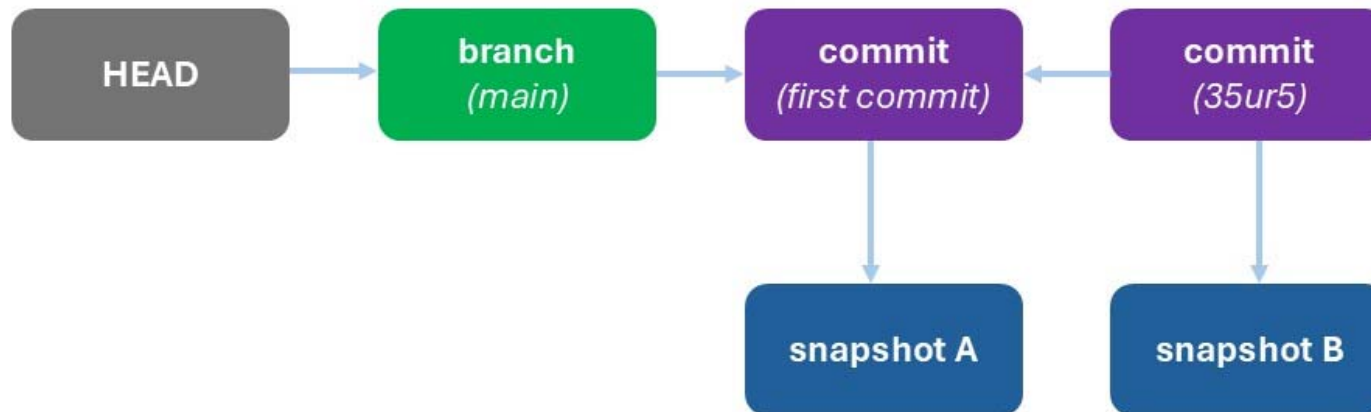
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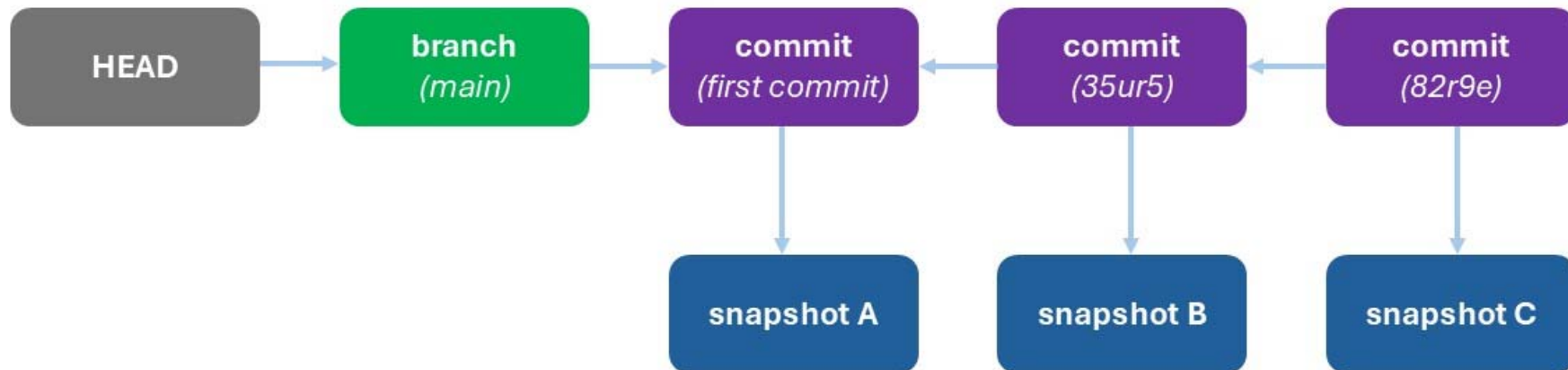
2.2 Committing and branching



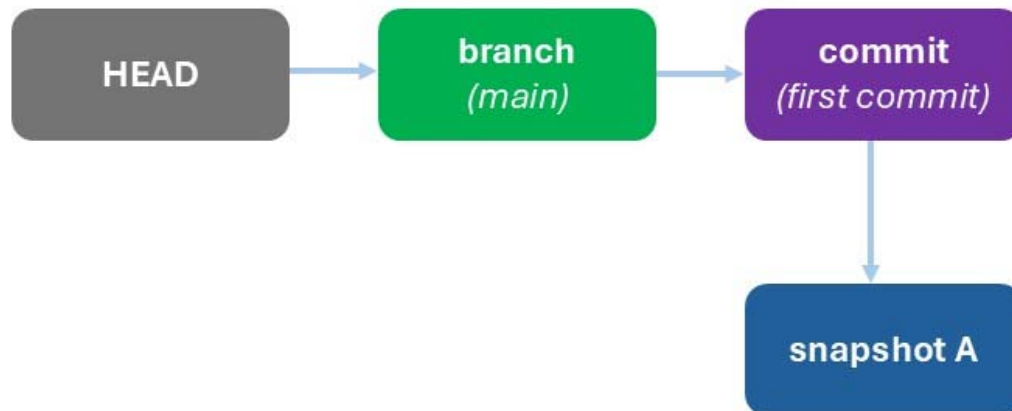
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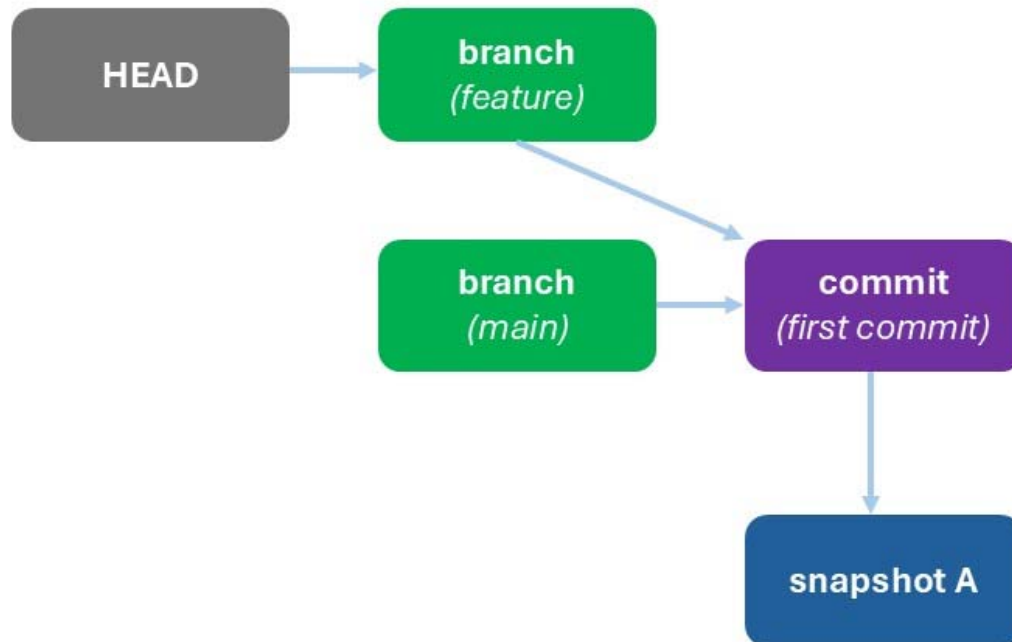
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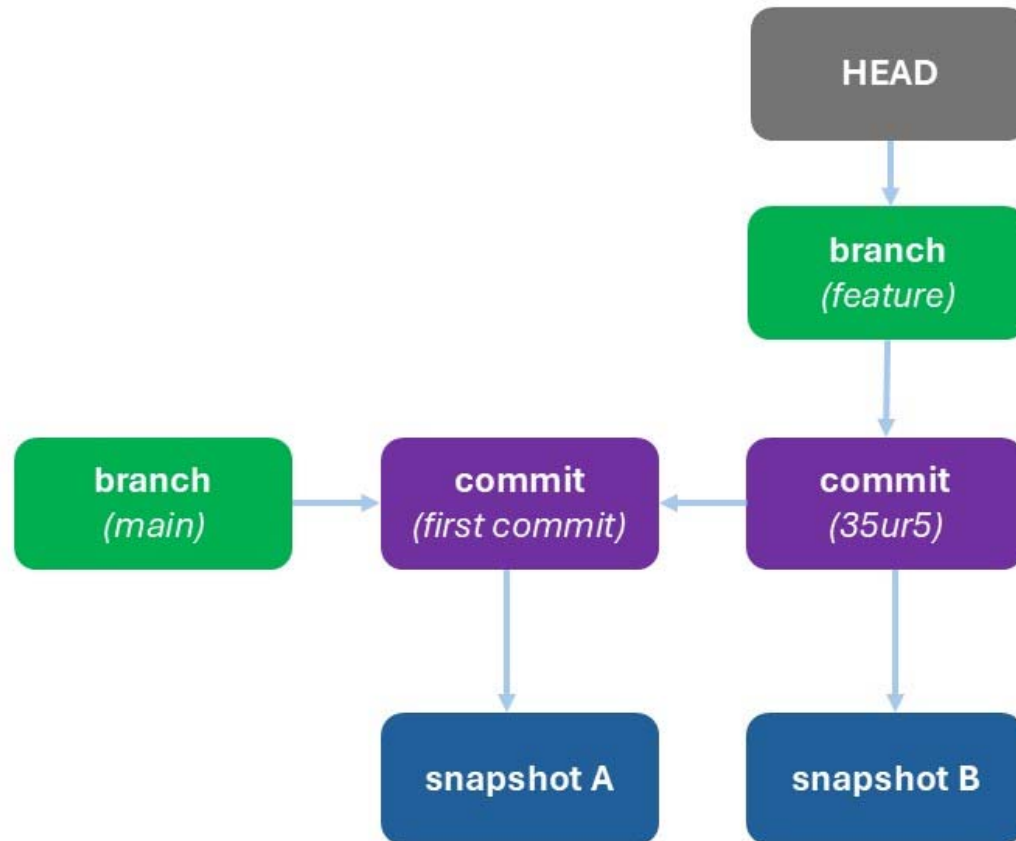
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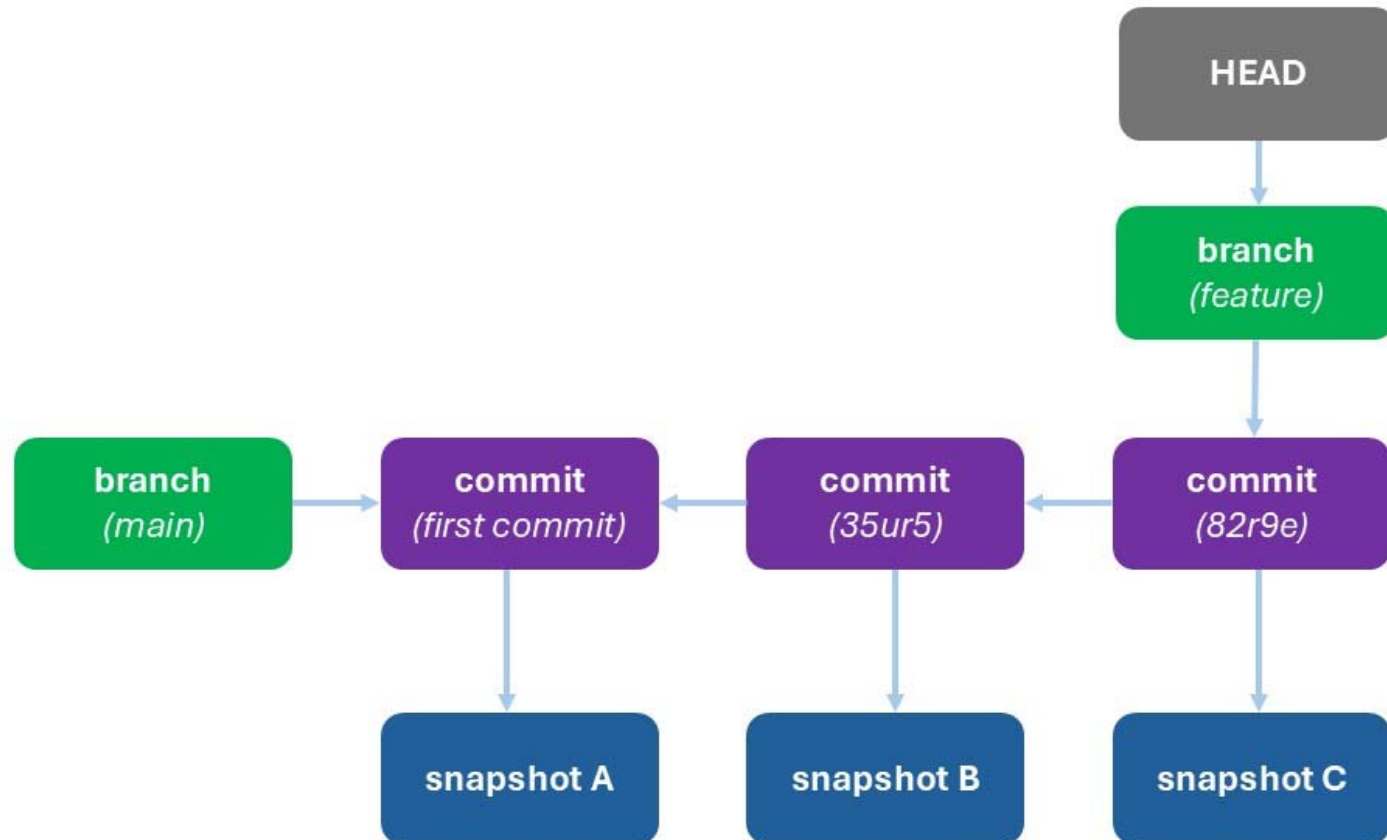
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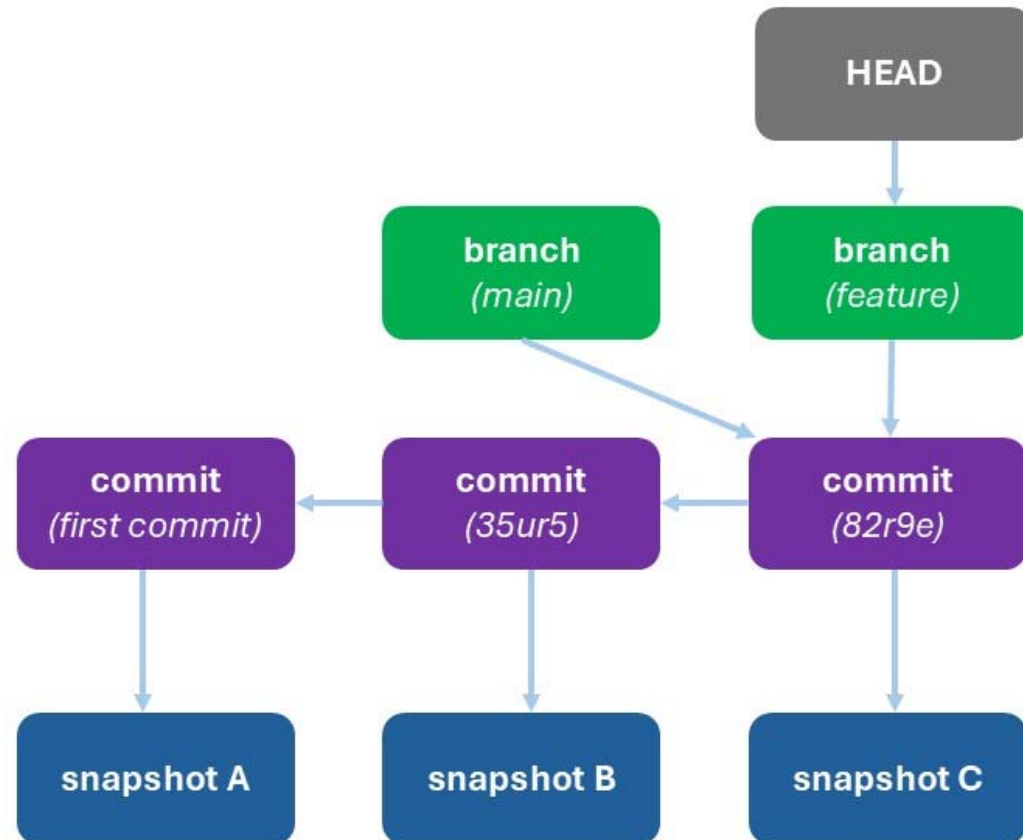
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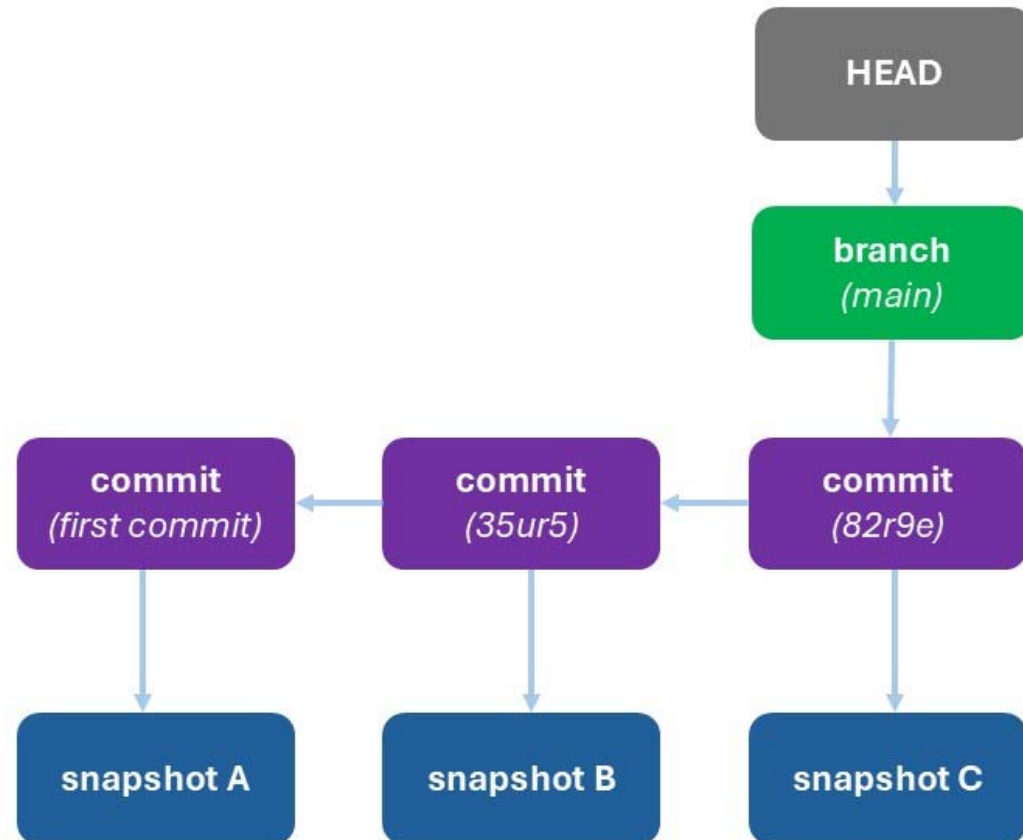
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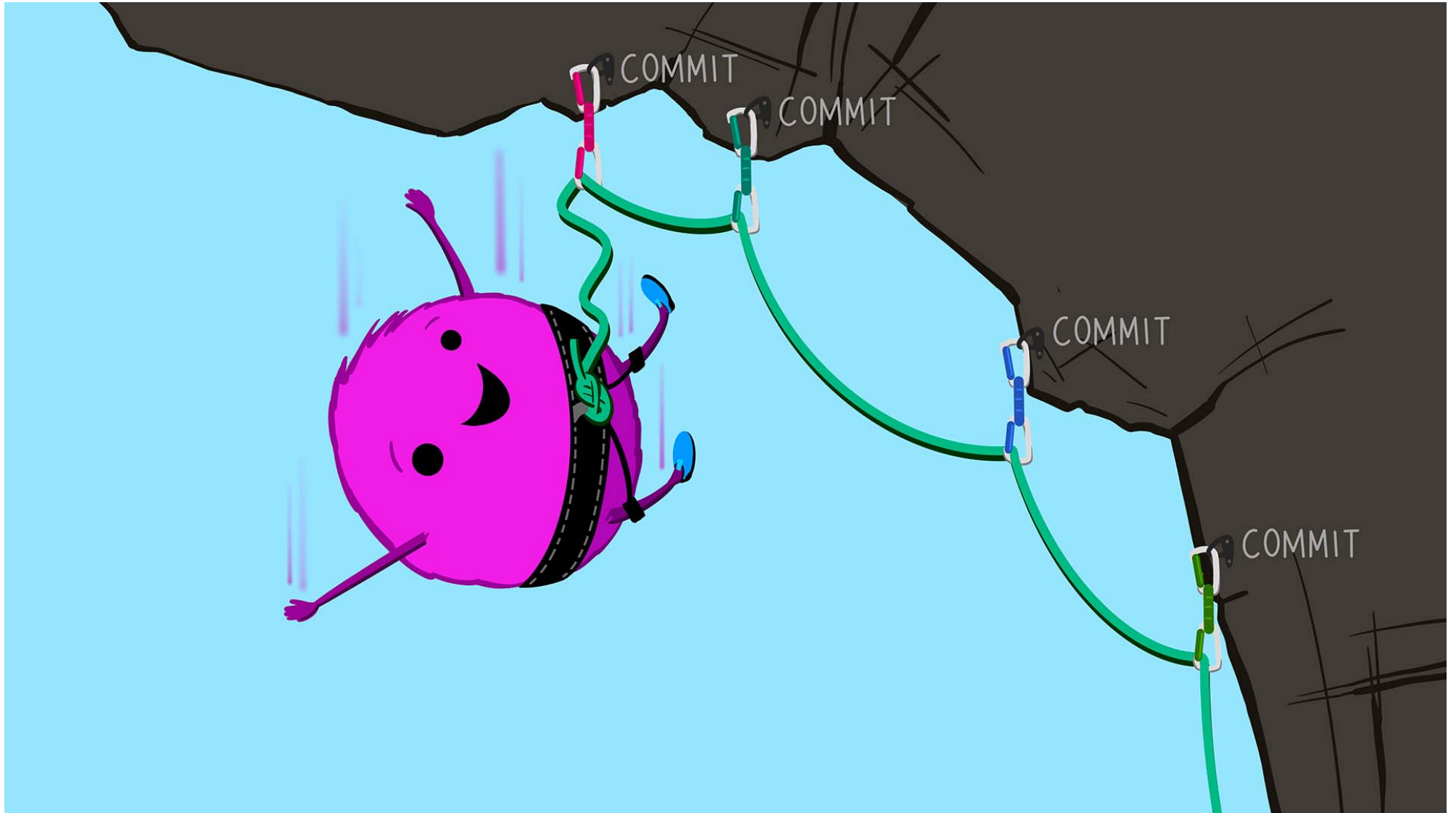
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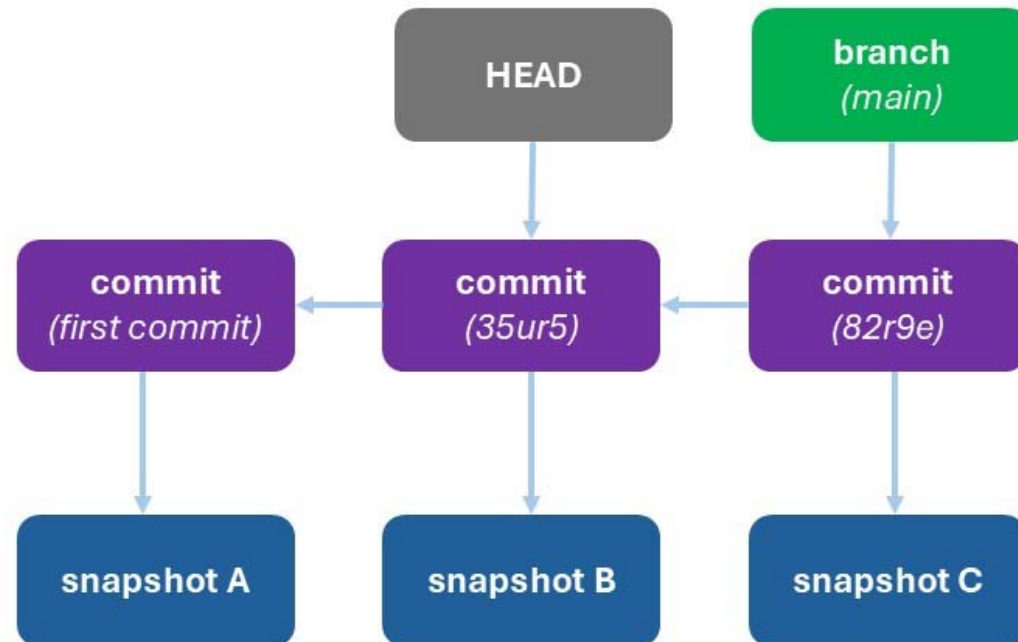
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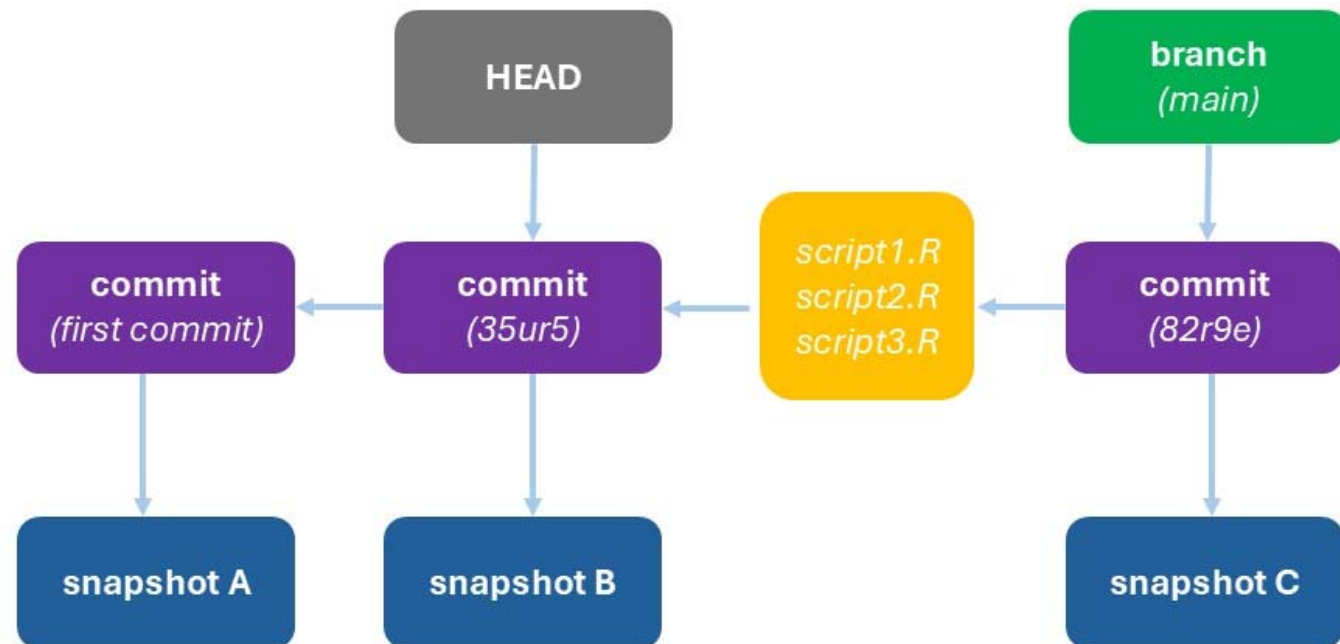
Better safe than sorry!



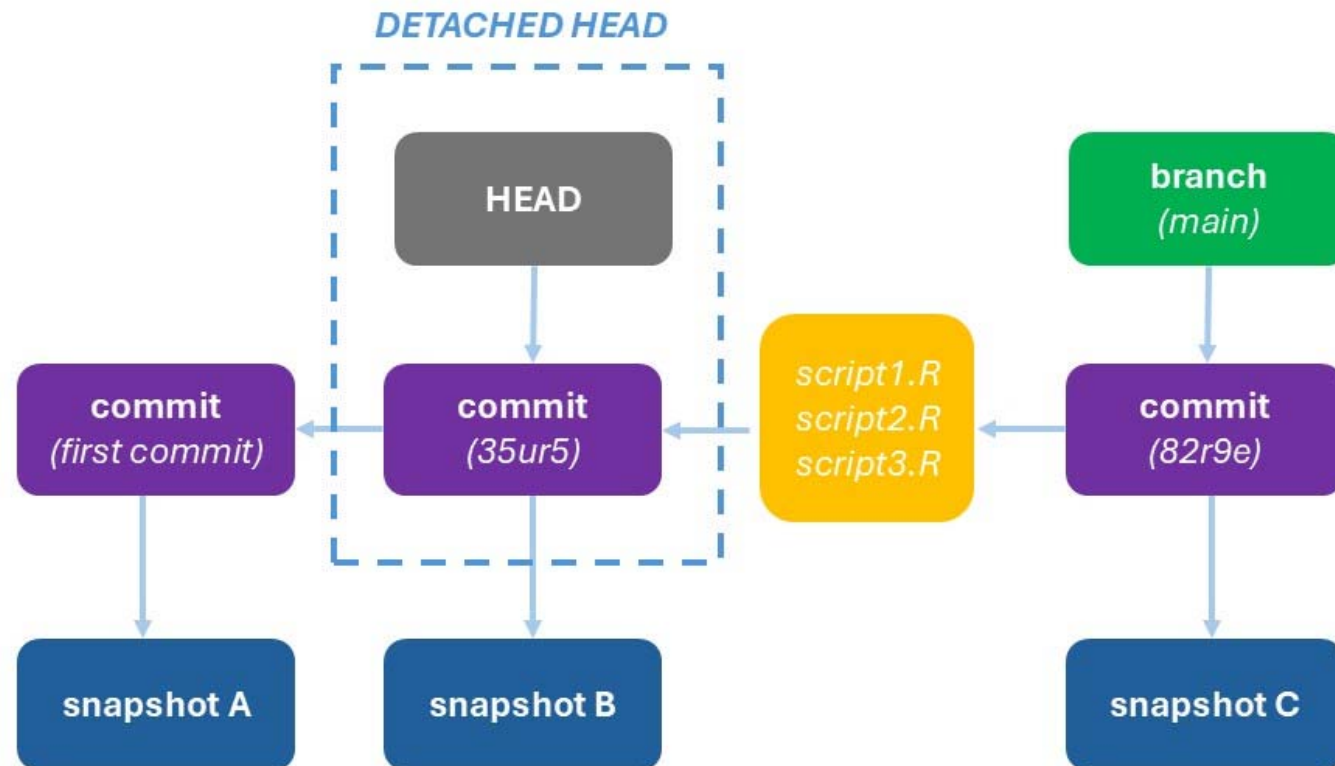
2.3 Detached head and commit



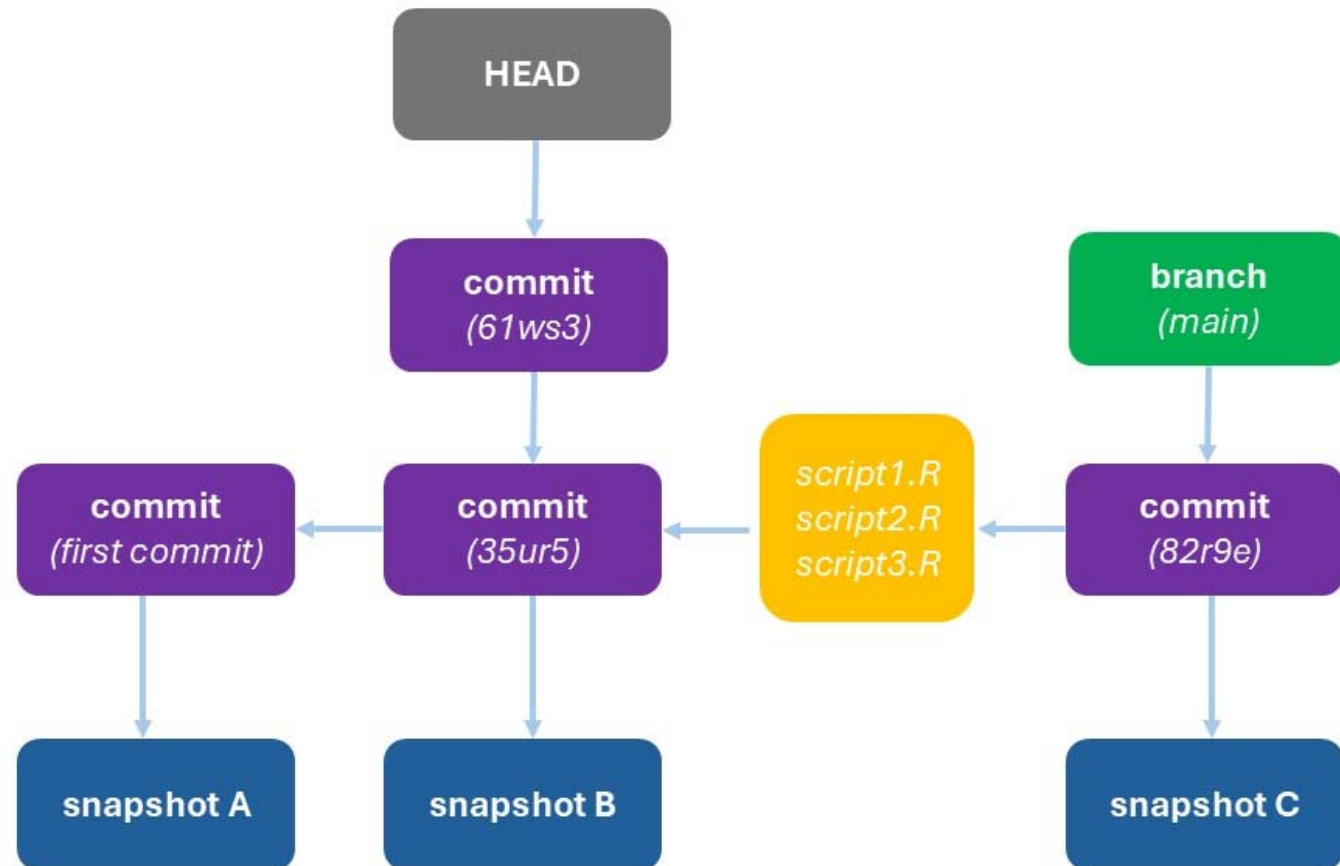
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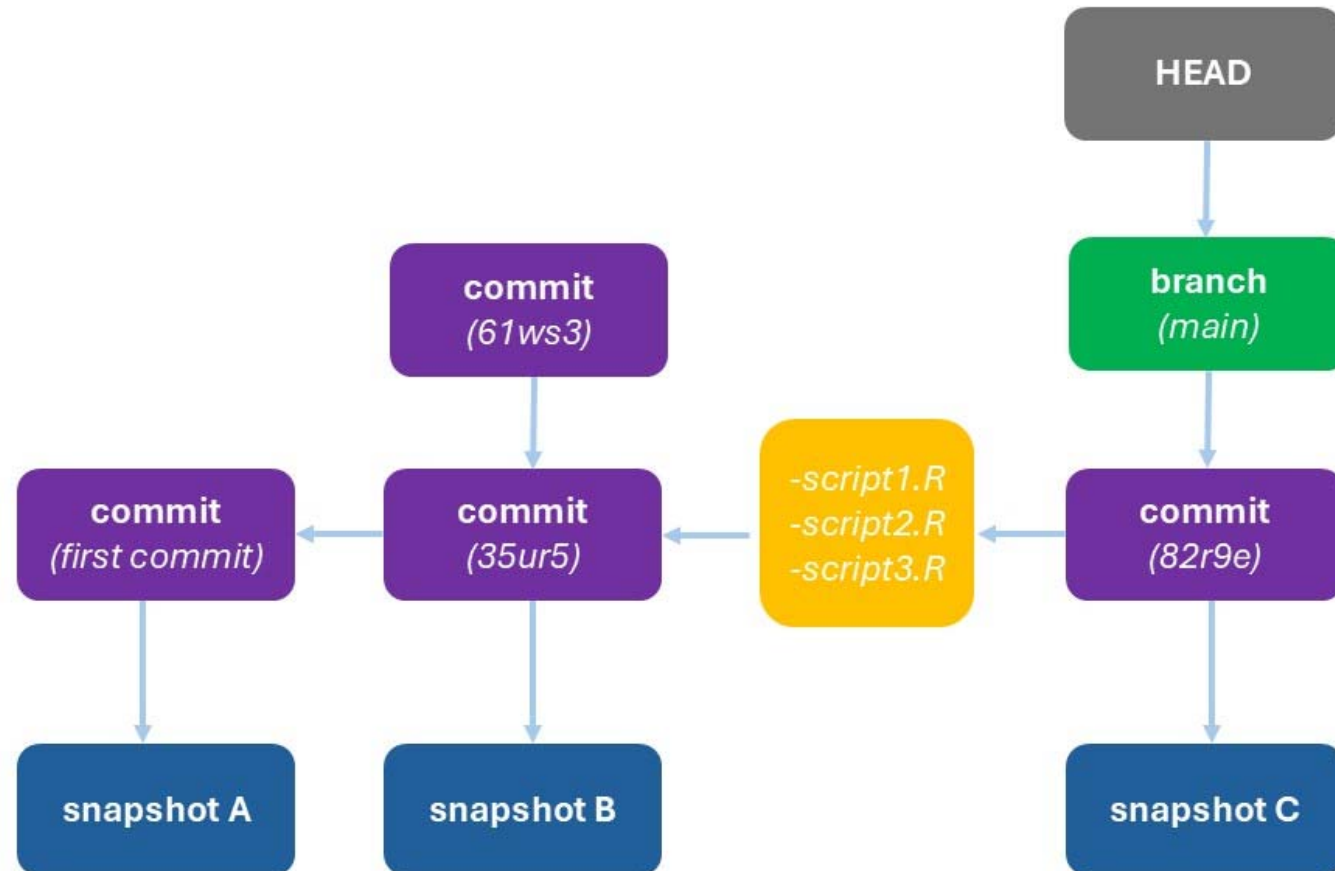
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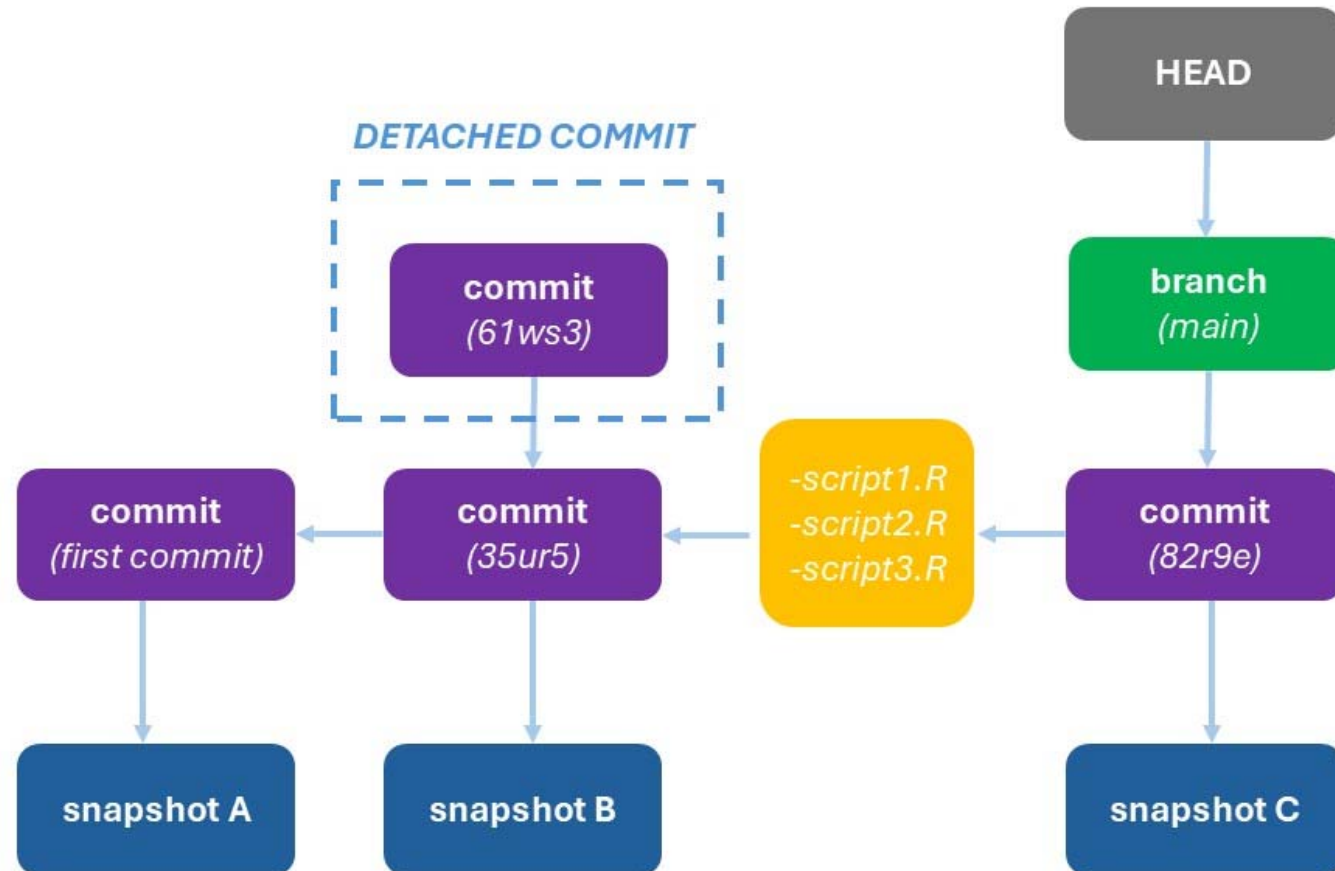
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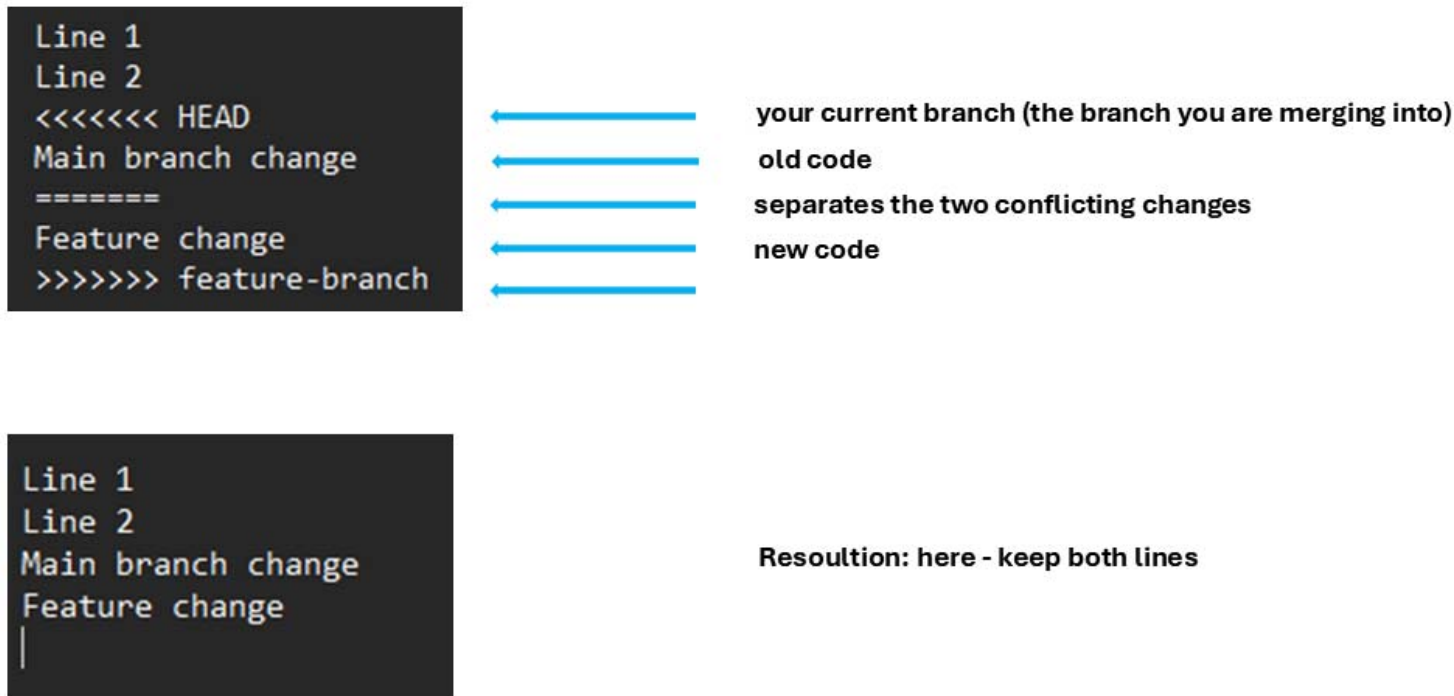
2.3 Detached head and commit



2.4 Merging and merge conflict

- **Merging:**
 - combines changes from one branch into another
 - typically integrating a feature branch into main
 - but it could also be a fix within a branch
- **Merge conflict:**
 - Git cannot automatically combine changes because the same part of a file was edited differently in both branches
- **Resolving conflicts:**
 - manually editing the conflicting files, keeping the desired changes, then staging and committing the resolution

2.4 Merging and merge conflict



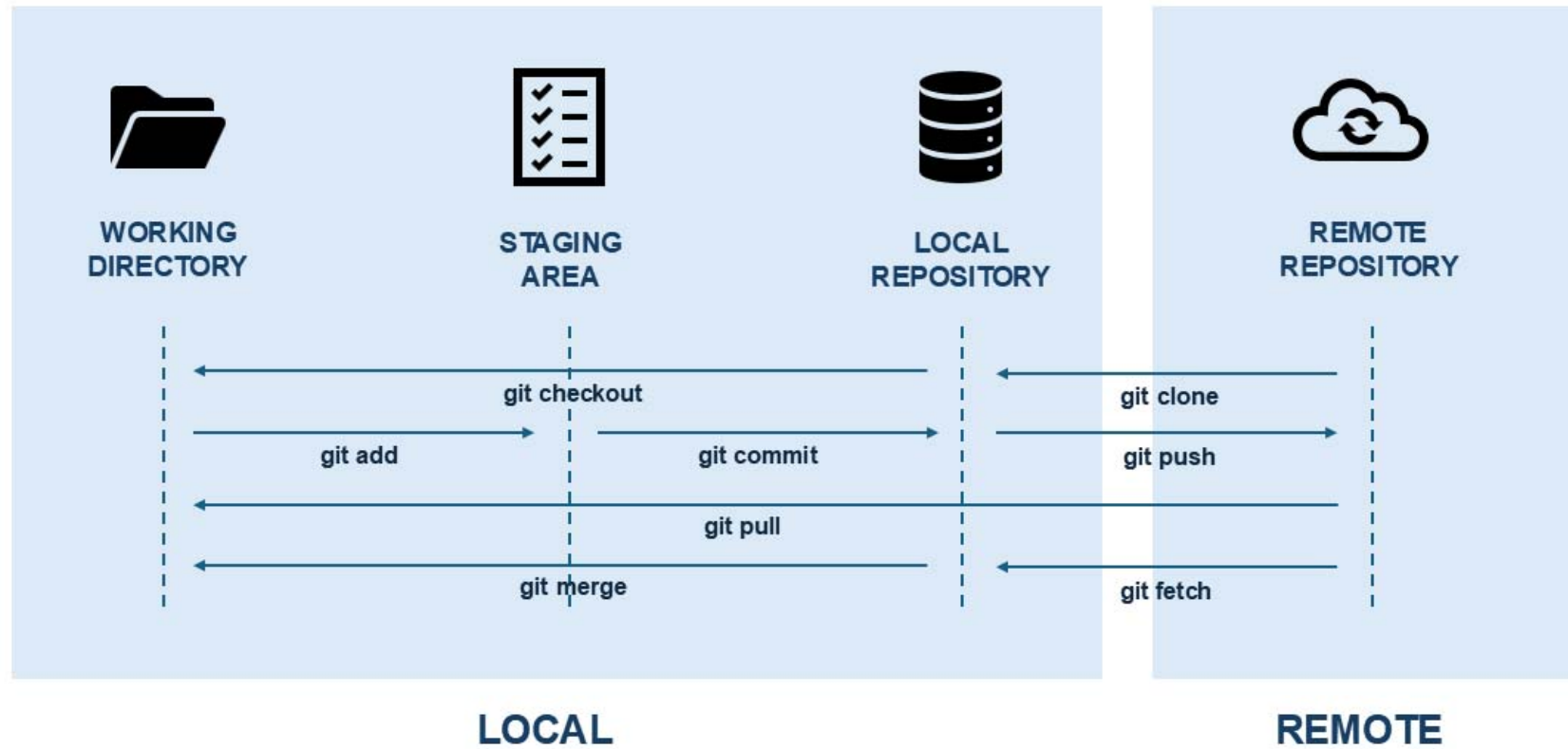
2.5 Locations



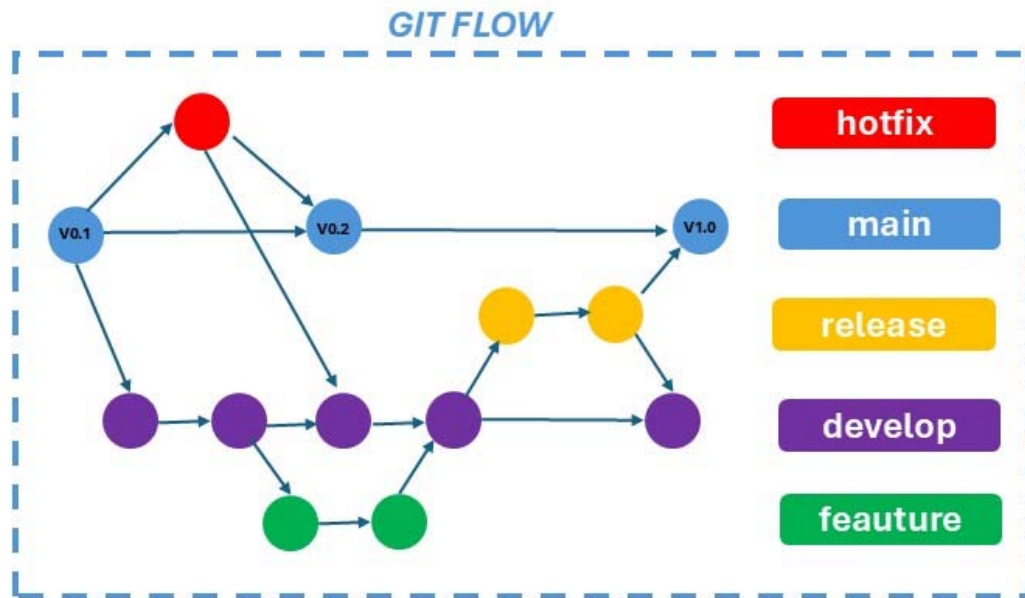
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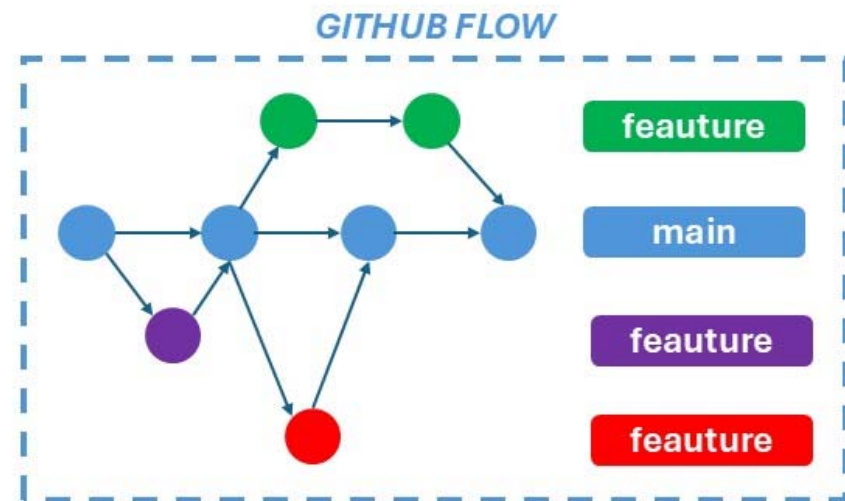
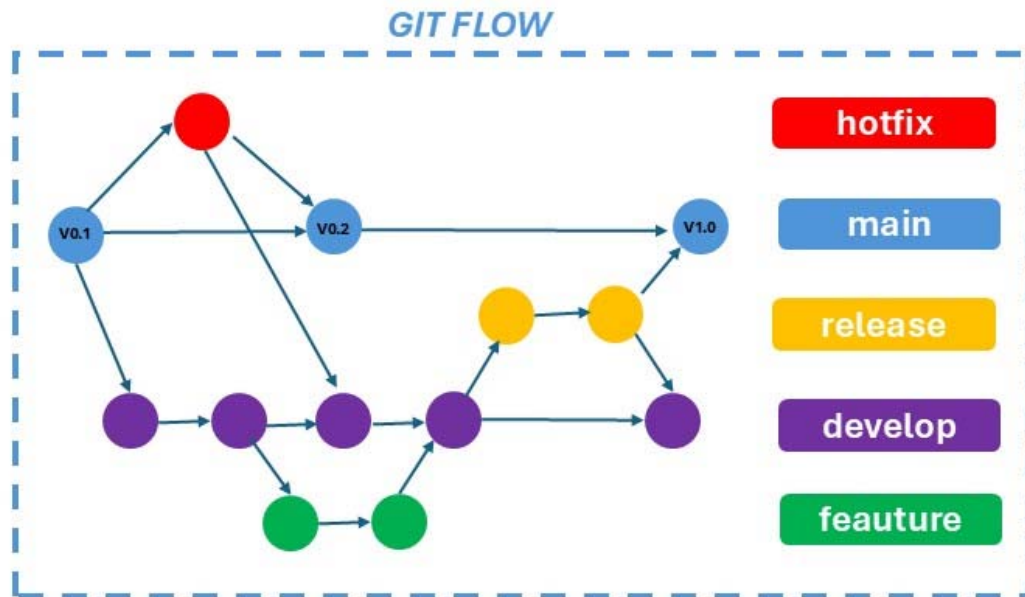
2.6 Git commands



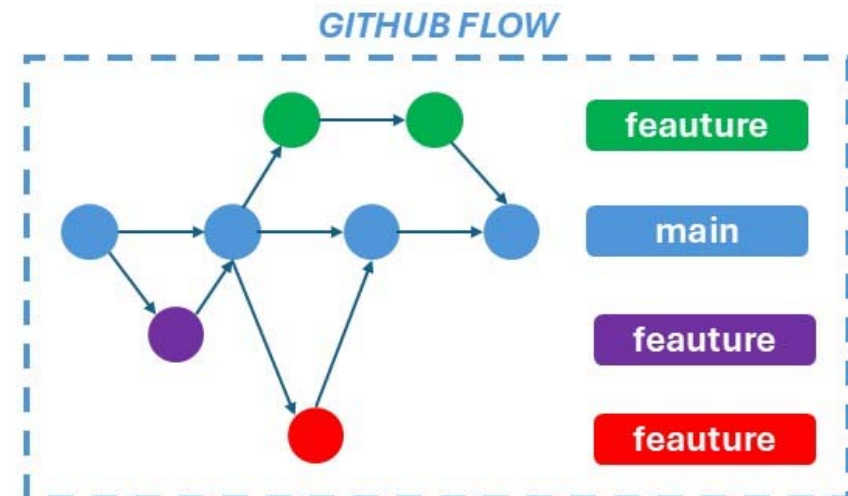
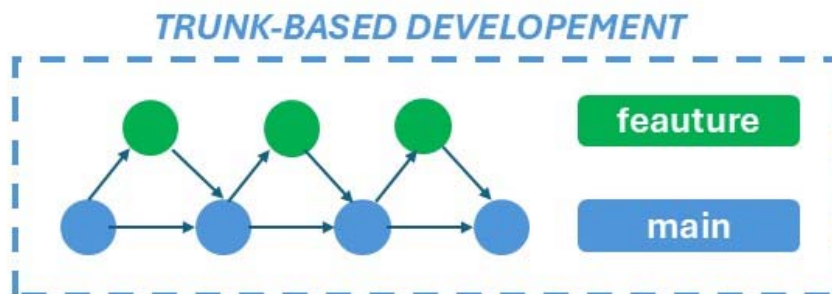
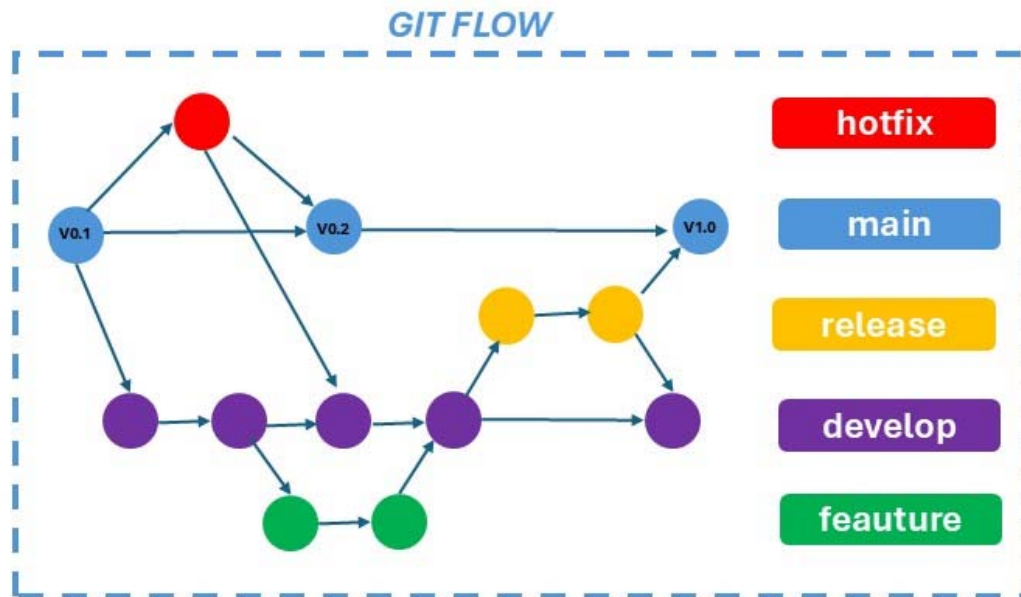
3 Collaboration strategy



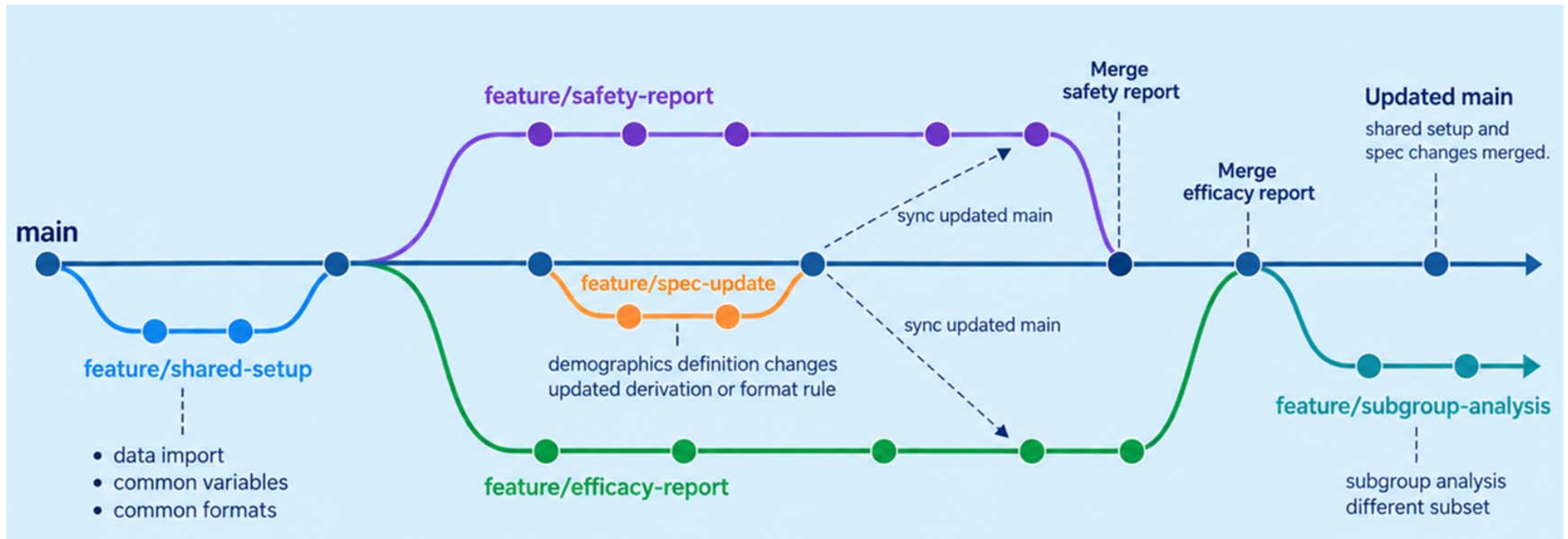
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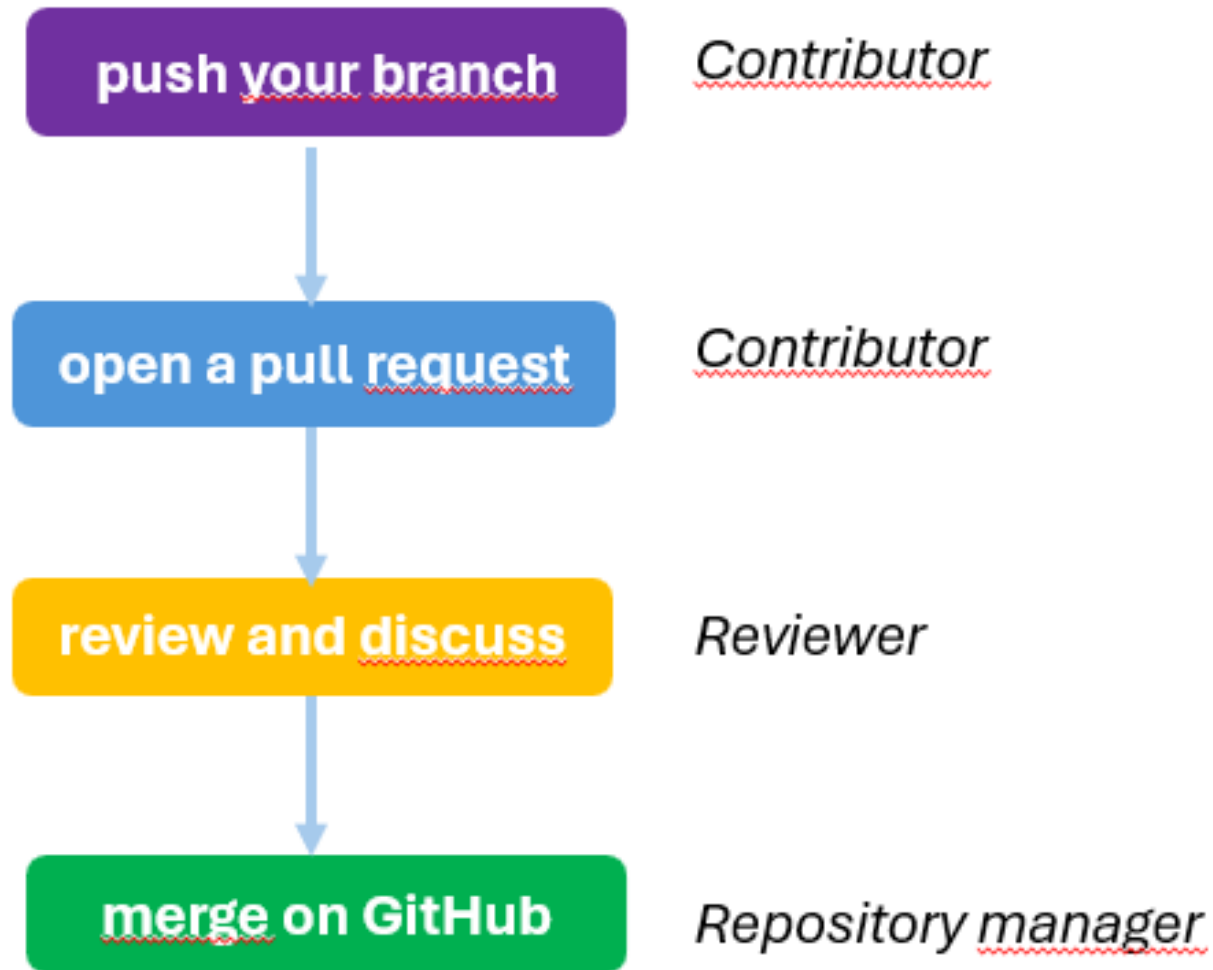


4 New SOP

4.1 Roles and responsibilities

- **Repository Manager**
 - creates and configures repositories
 - manages access & permissions
 - ensures that the main represents the current approved version
- **Contributor**
 - develops changes, submitting for review via pull request
 - addresses review comments, resolves merge conflicts affecting their branch
 - maintains local repository
- **Reviewer**
 - review of submitted code (proposed change is correct and appropriate)
 - requests changes or clarification where required, and
 - approves the pull request

4.2 Procedures



Thank you for you attention!

